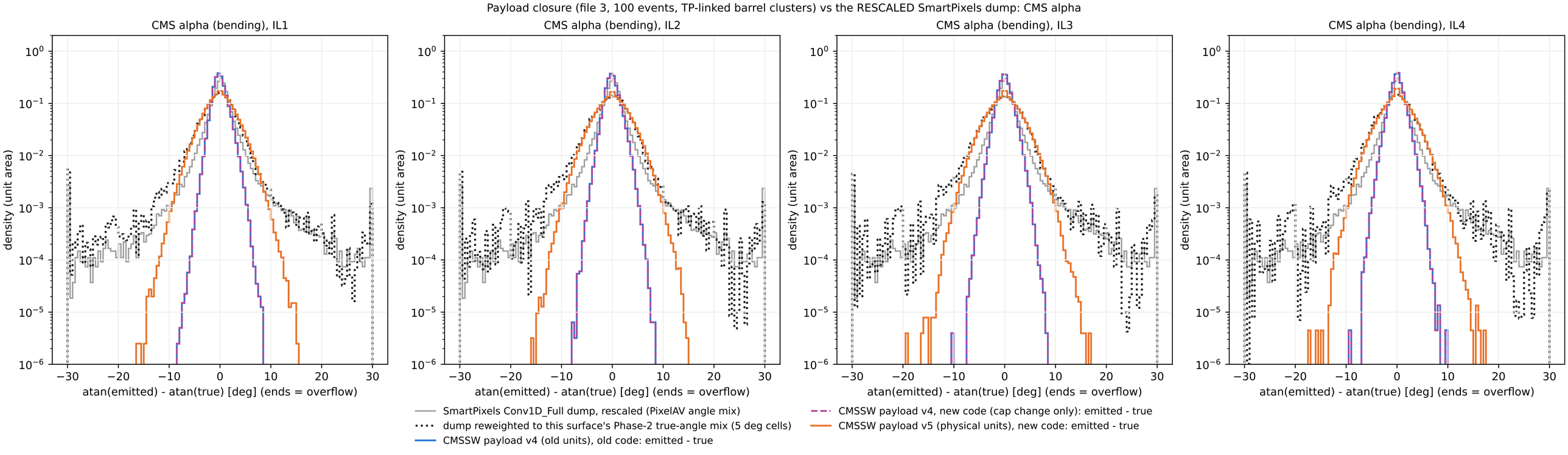
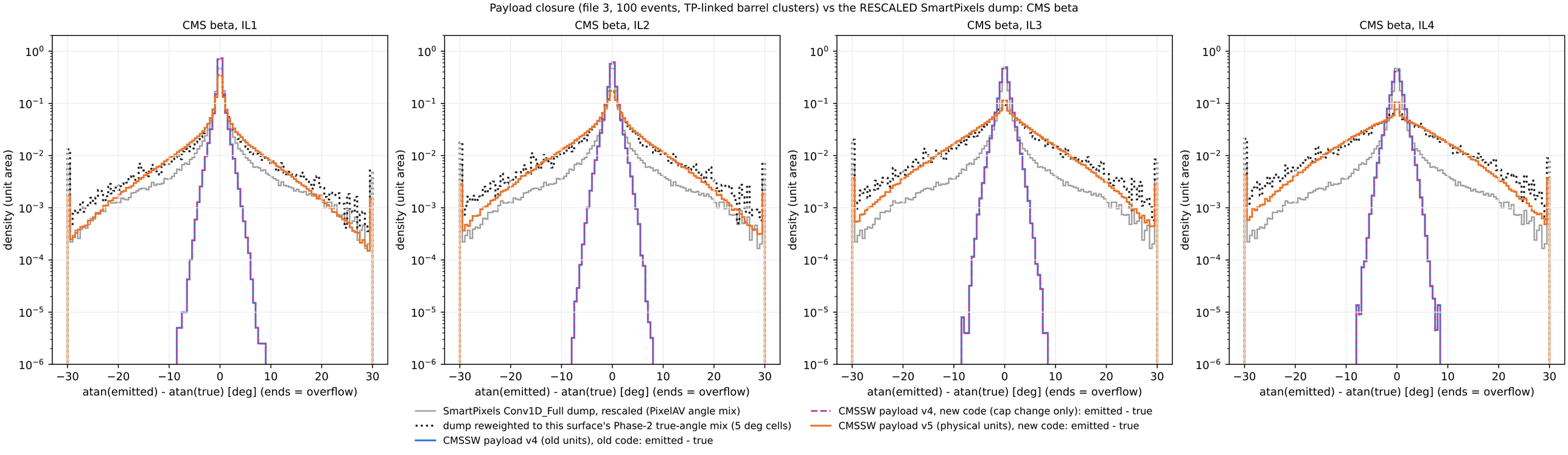

Payload closure (file 3, 100 events, TP-linked barrel clusters) vs the RESCALED SmartPixels dump: CMS alpha
CMS alpha (bending), IL1
CMS alpha (bending), IL2
CMS alpha (bending), IL3
CMS alpha (bending), IL4
density (unit area)
atan(emitted) - atan(true) [deg] (ends = overflow)
SmartPixels Conv1D_Full dump, rescaled (PixelAV angle mix)
dump reweighted to this surface's Phase-2 true-angle mix (5 deg cells)
CMSSW payload v4 (old units), old code: emitted - true
CMSSW payload v4, new code (cap change only): emitted - true
CMSSW payload v5 (physical units), new code: emitted - true


Payload closure (file 3, 100 events, TP-linked barrel clusters) vs the RESCALED SmartPixels dump: CMS beta
CMS beta, IL1
CMS beta, IL2
CMS beta, IL3
CMS beta, IL4
density (unit area)
atan(emitted) - atan(true) [deg] (ends = overflow)
SmartPixels Conv1D_Full dump, rescaled (PixelAV angle mix)
dump reweighted to this surface's Phase-2 true-angle mix (5 deg cells)
CMSSW payload v4 (old units), old code: emitted - true
CMSSW payload v4, new code (cap change only): emitted - true
CMSSW payload v5 (physical units), new code: emitted - true

Payload closure statistics [deg]: core RMS (mode-seeded Gaussian core, about 0), full RMS (about 0), fraction beyond 10 deg; 'covered' = share of the surface's clusters whose (alpha, beta) 5-deg cell has dump rows (1)

angle	surface	curve	N	core_rms_deg	full_rms_deg	frac_abs_gt10deg	phase2_cells_covered_by_dump
alpha	IL1	SmartPixels Conv1D_Full dump, rescaled (PixelAV angle mix)	108222	1.273	3.783	0.0157	
alpha	IL1	dump reweighted to this surface's Phase-2 true-angle mix (5 deg cells)	108222	2.421	4.431	0.02451	0.9745
alpha	IL1	CMSSW payload v4 (old units), old code: emitted - true	800939	1.05	1.251	0	
alpha	IL1	CMSSW payload v4, new code (cap change only): emitted - true	803522	1.046	1.249	0	
alpha	IL1	CMSSW payload v5 (physical units), new code: emitted - true	803522	2.49	2.788	0.001462	
alpha	IL2	SmartPixels Conv1D_Full dump, rescaled (PixelAV angle mix)	108222	1.273	3.783	0.0157	
alpha	IL2	dump reweighted to this surface's Phase-2 true-angle mix (5 deg cells)	108222	2.842	4.638	0.02812	0.9627
alpha	IL2	CMSSW payload v4 (old units), old code: emitted - true	625250	1.071	1.286	0	
alpha	IL2	CMSSW payload v4, new code (cap change only): emitted - true	628154	1.064	1.283	0	
alpha	IL2	CMSSW payload v5 (physical units), new code: emitted - true	628154	2.774	3.016	0.002205	
alpha	IL3	SmartPixels Conv1D_Full dump, rescaled (PixelAV angle mix)	108222	1.273	3.783	0.0157	
alpha	IL3	dump reweighted to this surface's Phase-2 true-angle mix (5 deg cells)	108222	2.926	4.713	0.0276	0.9495
alpha	IL3	CMSSW payload v4 (old units), old code: emitted - true	504458	1.093	1.309	1.982e-06	
alpha	IL3	CMSSW payload v4, new code (cap change only): emitted - true	507823	1.083	1.305	1.969e-06	
alpha	IL3	CMSSW payload v5 (physical units), new code: emitted - true	507823	2.773	3.016	0.002148	
alpha	IL4	SmartPixels Conv1D_Full dump, rescaled (PixelAV angle mix)	108222	1.273	3.783	0.0157	
alpha	IL4	dump reweighted to this surface's Phase-2 true-angle mix (5 deg cells)	108222	2.867	4.788	0.02606	0.9323
alpha	IL4	CMSSW payload v4 (old units), old code: emitted - true	443469	1.058	1.304	0	
alpha	IL4	CMSSW payload v4, new code (cap change only): emitted - true	447274	1.044	1.298	0	
alpha	IL4	CMSSW payload v5 (physical units), new code: emitted - true	447274	2.579	2.916	0.001874	
beta	IL1	SmartPixels Conv1D_Full dump, rescaled (PixelAV angle mix)	108222	0.4863	6.407	0.06576	
beta	IL1	dump reweighted to this surface's Phase-2 true-angle mix (5 deg cells)	108222	0.5374	8.199	0.1264	0.9745
beta	IL1	CMSSW payload v4 (old units), old code: emitted - true	797349	0.126	0.7665	0	
beta	IL1	CMSSW payload v4, new code (cap change only): emitted - true	803522	0.1205	0.7635	0	
beta	IL1	CMSSW payload v5 (physical units), new code: emitted - true	803522	0.5098	5.842	0.09643	
beta	IL2	SmartPixels Conv1D_Full dump, rescaled (PixelAV angle mix)	108222	0.4863	6.407	0.06576	
beta	IL2	dump reweighted to this surface's Phase-2 true-angle mix (5 deg cells)	108222	1.904	9.615	0.1829	0.9627
beta	IL2	CMSSW payload v4 (old units), old code: emitted - true	620380	0.4474	0.9467	0	
beta	IL2	CMSSW payload v4, new code (cap change only): emitted - true	628154	0.4293	0.9408	0	
beta	IL2	CMSSW payload v5 (physical units), new code: emitted - true	628154	2.165	7.113	0.1447	
beta	IL3	SmartPixels Conv1D_Full dump, rescaled (PixelAV angle mix)	108222	0.4863	6.407	0.06576	
beta	IL3	dump reweighted to this surface's Phase-2 true-angle mix (5 deg cells)	108222	4.642	10.6	0.2399	0.9495
beta	IL3	CMSSW payload v4 (old units), old code: emitted - true	499080	0.7213	1.101	0	
beta	IL3	CMSSW payload v4, new code (cap change only): emitted - true	507823	0.6985	1.091	0	

Payload closure statistics [deg]: core RMS (mode-seeded Gaussian core, about 0), full RMS (about 0), fraction beyond 10 deg; 'covered' = share of the surface's clusters whose (alpha, beta) 5-deg cell has dump rows (2)

angle	surface	curve	N	core_rms_deg	full_rms_deg	frac_abs_gt10deg	phase2_cells_covered_by_dump
beta	IL3	CMSSW payload v5 (physical units), new code: emitted - true	507823	5.426	8.211	0.1949	
beta	IL4	SmartPixels Conv1D_Full dump, rescaled (PixelAV angle mix)	108222	0.4863	6.407	0.06576	
beta	IL4	dump reweighted to this surface's Phase-2 true-angle mix (5 deg cells)	108222	6.446	11.13	0.2798	0.9323
beta	IL4	CMSSW payload v4 (old units), old code: emitted - true	438400	0.8697	1.186	0	
beta	IL4	CMSSW payload v4, new code (cap change only): emitted - true	447274	0.8391	1.174	0	
beta	IL4	CMSSW payload v5 (physical units), new code: emitted - true	447274	6.882	8.803	0.2278	

Payload residuals in cot (the payload's own quantity), TP-linked barrel clusters: q-width, RMS, the published sigma (median) and the pull q-width (residual / published sigma)

surface	run	angle	n	q_width_cot	rms_cot	median_sigma	pull_q_width
IL1	payload v4 (old units), old code	alpha	800939	0.02136	0.02325	0.01924	1.107
IL1	payload v4 (old units), old code	beta	797349	0.02724	0.02624	0.01956	1.452
IL1	payload v4, new code (cap change only)	alpha	803522	0.02136	0.02325	0.01924	1.106
IL1	payload v4, new code (cap change only)	beta	803522	0.02722	0.02623	0.01956	1.453
IL1	payload v5 (physical units), new code	alpha	803522	0.04907	0.05173	0.05377	1.005
IL1	payload v5 (physical units), new code	beta	803522	0.1729	0.1782	0.1685	1.014
IL2	payload v4 (old units), old code	alpha	625250	0.02204	0.02393	0.01953	1.079
IL2	payload v4 (old units), old code	beta	620380	0.02682	0.02621	0.02214	1.34
IL2	payload v4, new code (cap change only)	alpha	628154	0.02204	0.02393	0.01953	1.079
IL2	payload v4, new code (cap change only)	beta	628154	0.02686	0.02628	0.02203	1.343
IL2	payload v5 (physical units), new code	alpha	628154	0.05414	0.05586	0.05585	1.007
IL2	payload v5 (physical units), new code	beta	628154	0.1841	0.189	0.1835	1.017
IL3	payload v4 (old units), old code	alpha	504458	0.02321	0.02489	0.02347	1.056
IL3	payload v4 (old units), old code	beta	499080	0.02631	0.02619	0.02306	1.22
IL3	payload v4, new code (cap change only)	alpha	507823	0.02319	0.02488	0.02347	1.057
IL3	payload v4, new code (cap change only)	beta	507823	0.02634	0.02634	0.02306	1.222
IL3	payload v5 (physical units), new code	alpha	507823	0.05605	0.05732	0.05585	1.005
IL3	payload v5 (physical units), new code	beta	507823	0.1929	0.1967	0.2014	1.026
IL4	payload v4 (old units), old code	alpha	443469	0.02399	0.02554	0.02425	1.041
IL4	payload v4 (old units), old code	beta	438400	0.02555	0.02589	0.02322	1.133
IL4	payload v4, new code (cap change only)	alpha	447274	0.02395	0.02553	0.02425	1.041
IL4	payload v4, new code (cap change only)	beta	447274	0.02557	0.02589	0.02322	1.137
IL4	payload v5 (physical units), new code	alpha	447274	0.05621	0.05738	0.05585	1.008
IL4	payload v5 (physical units), new code	beta	447274	0.1923	0.1961	0.1959	1.027

hasAlpha / hasBeta, TP-linked (tpKey >= 0) vs unlinked, and the share of reported |cot| > 12

run	group	linked	n	hasAlpha	hasBeta	frac_abs_cotA_gt12	frac_abs_cotB_gt12
payload v4 (old units), old code	IL2-4	True	1583251	0.9936	0.984	0	0
payload v4 (old units), old code	IL2-4	False	231810	1	1	0	0
payload v4 (old units), old code	IL1	True	803522	0.9968	0.9923	0	0
payload v4 (old units), old code	IL1	False	95569	1	1	0	0
payload v4 (old units), old code	ID1-4	True	3706158	0	0	nan	nan
payload v4 (old units), old code	ID1-4	False	826596	0	0	nan	nan
payload v4, new code (cap change only)	IL2-4	True	1583251	1	1	0.006363	0.01604
payload v4, new code (cap change only)	IL2-4	False	231810	1	1	0	0
payload v4, new code (cap change only)	IL1	True	803522	1	1	0.003213	0.007682
payload v4, new code (cap change only)	IL1	False	95569	1	1	0	0
payload v4, new code (cap change only)	ID1-4	True	3706158	0	0	nan	nan
payload v4, new code (cap change only)	ID1-4	False	826596	0	0	nan	nan
payload v5 (physical units), new code	IL2-4	True	1583251	1	1	0.006357	0.01602
payload v5 (physical units), new code	IL2-4	False	231810	1	1	0	0
payload v5 (physical units), new code	IL1	True	803522	1	1	0.003208	0.00767
payload v5 (physical units), new code	IL1	False	95569	1	1	0	0
payload v5 (physical units), new code	ID1-4	True	3706158	0	0	nan	nan
payload v5 (physical units), new code	ID1-4	False	826596	0	0	nan	nan

The cap change alone (payload v4, old code vs new code)

check	result
payload_cap_change_detail	{"same_rows": true, "cot_identical": true, "hasAlpha_changed": 12657, "hasBeta_changed": 31564, "hasAlpha_changed_and_abs_cot_gt12": 12656, "hasBeta_changed_and_abs_cot_gt12": 31563}
payload_v4_oldcode_vs_newcode_SP	{"n_common_events": 100, "n_branches_compared": 245, "n_identical": 243, "differing": ["L1TSmartPixelsCluster_hasAlpha", "L1TSmartPixelsCluster_hasBeta"], "only_in_one": []}
payload_v4_oldcode_vs_newcode_nonSP	{"n_common_events": 100, "n_branches_compared": 617, "n_identical": 617, "differing": [], "only_in_one": []}